

Total No. of Questions : 10]

SEAT No. :

P1871

[4859]-1060

[Total No. of Pages : 3

B.E. (I.T.)

MACHINE LEARNING

(2012 Course) (414455) (End Semester) (Semester-I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Draw neat diagrams wherever necessary.*
- 2) *Assume suitable data, if necessary.*
- 3) *Figures to the right indicate full marks.*

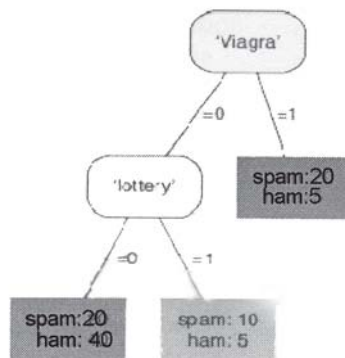
Q1) a) Explain predictive and descriptive tasks. **[5]**

b) Prove with an example **Accuracy = 1-error rate.** **[5]**

OR

Q2) a) Define class probability estimator. State mathematical model of class probability estimator. Is that a predictive or descriptive task? Justify. **[5]**

b) What is majority class decision rule? Using following feature tree, write decision rules for majority class. **[5]**



Q3) a) What is a slack variable? Discuss margin errors. **[5]**

b) Explain ridge regression and lasso. **[5]**

OR

Q4) a) Consider the following three-class confusion matrix. **[5]**

P.T.O.

Predicted			
Actual	15	2	3
	7	15	8
	2	3	45

Calculate precision and recall per class. Also calculate weighted average precision and recall for the classifier.

- b) Explain the term bias-variance dilemma. [5]

Q5) a) Explain with the help of diagrams and equations Minkowski, Euclidean, Manhattan and Hamming distances. [8]

- b) What is a feature tree? Write the Grow Tree algorithm to generate feature tree. Explain the role of best split in this algorithm. [10]

OR

Q6) a) Explain support and confidence with the help of formulae. Calculate support, and confidence for the following example. [8]

Transaction	Items
1	nappies
2	beer, crisps
3	apples, nappies
4	beer, crisps, nappies
5	apples
6	apples, beer, crisps, nappies
7	apples, crisps
8	crisps

- b) Write an algorithm for K-means clustering. Describe its working in brief using example. [10]

Q7) a) Distinguish between discriminative learning models and generative learning model with suitable examples. [8]

b) Define: [8]

- 1) Bernoulli's distribution.
- 2) Binomial distribution.
- 3) MAP decision rule.
- 4) Maximum likelihood function.

OR

Q8) a) Write a note on Naïve Bayes Classification algorithm. [8]

b) Explain in brief logistic regression. Compare simple regression and logistic regression. [8]

Q9) a) Explain reinforcement learning. [8]

b) Explain bagging and boosting as ensemble methods. [8]

OR

Q10)a) Explain data stream and online learning. [8]

b) Explain multitask learning. [8]

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